

KORMARINE CONFERENCE 2026 · 1 SEPTEMBER 2026 · WYNDHAM GRAND HOTEL, BUSAN, KOREA

SESSION K1-3 ENVIRONMENTAL REGULATIONS IN SHIPPING & SHIPBUILDING

CONFERENCE THEME · NAVIGATING UNCERTAINTY: GEOPOLITICS, DECARBONIZATION AND THE FUTURE OF MARITIME INDUSTRY

Beyond 2030

Innovation-Driven Strategies for Carbon-Neutral Shipping and Ports

Innovation Reclassified — From the Supply Readiness Deficit to Interface Sovereignty

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Data verified against primary sources · 23 August 2026

CONTEXT · This talk rests on three studies: a 45-expert Delphi (Ahn 2023), the SRD framework (Ahn 2026) and Jeon et al. (2025) — all Korea Maritime Institute.

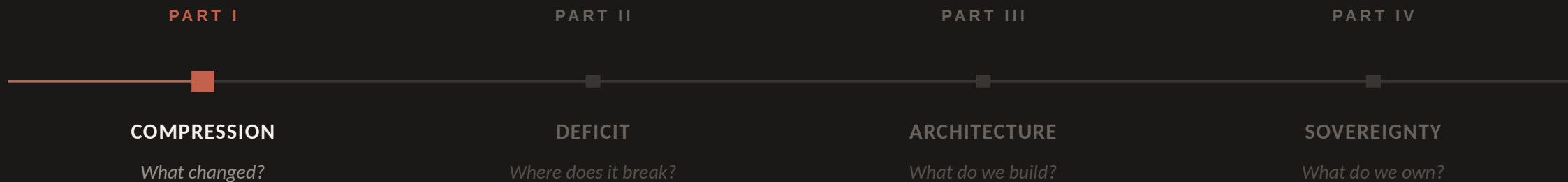
 [F] VERIFIED FACT  [A] AUTHOR CALC.  [M] MODEL OUTPUT  [E] EXPERT PANEL  [H] HYPOTHESIS



The Compression of Time

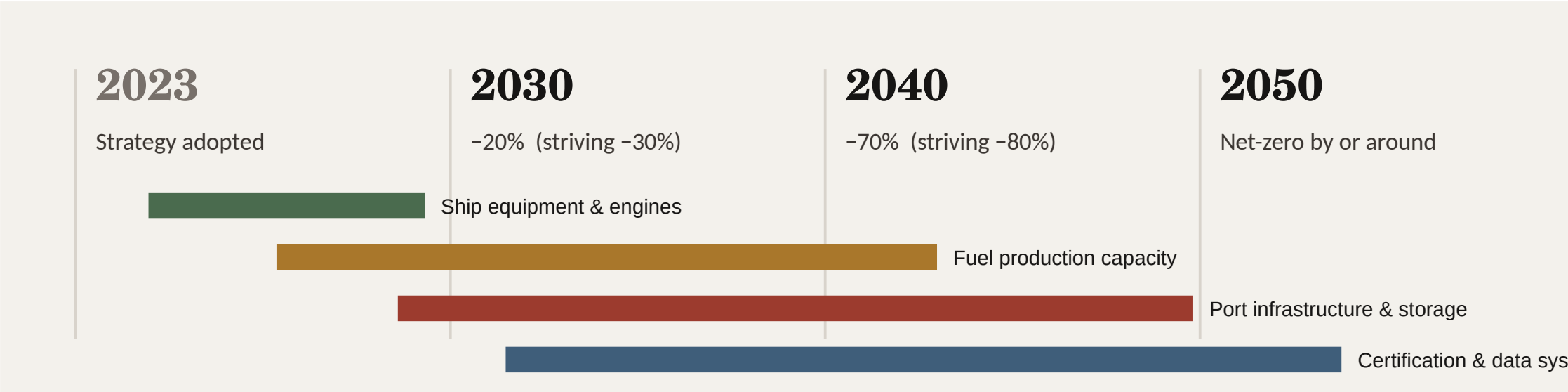
Innovation has already been reclassified. What binds us now is not the height of the target but the interval between targets.

- IEA The reclassification reaches shipping in March 2026
- IMO Targets imply a 3.7–5.5× acceleration after 2030
- DNV 50, 25 and 1 Mtoe are three different numbers



2030 Is a Checkpoint, Not the Finish Line

The risk is not the height of the target — it is the interval between targets.



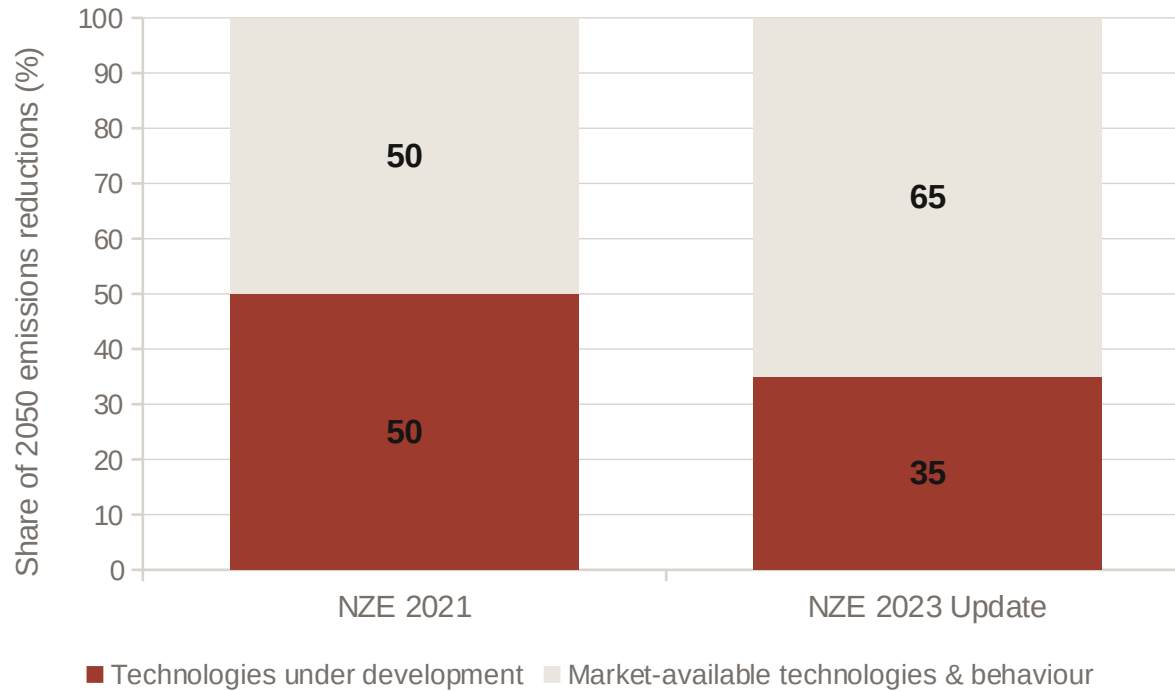
TIME-TO-BUILD vs TIME-TO-COMPLY

[H] The policy failure this study targets is not the absence of near-term technology. It is a sequencing failure: long-lead complementary assets and interoperability rules are built last, when they must be built first.

CONTEXT • 22 years bought 20–30 points; the next 50 must arrive in 10 — and the slowest assets to build, certification and data systems, are the ones started last.

The IEA Innovation Shift: Progress, Not Illusion

The fall from ~50% to ~35% is genuine innovation progress — not a statistical artefact.



[F] IEA 2021

Nearly half of 2050 reductions came from technologies then at demonstration or prototype stage — higher still in heavy industry and long-distance transport.

[F] IEA 2023

That share fell to around 35%. The IEA attributes the fall to innovation in battery chemistries and to the market expansion of commercial technologies such as solar PV.

[H] Reading

The bottleneck did not disappear. It migrated — from individual devices to fuel production, infrastructure, safety, verification and interoperability.

CONTEXT • The IEA credits the fall to real innovation — better batteries, cheaper solar; in 2021 the under-development share was higher still in shipping's own column.

The two scenarios do not share base year, technology taxonomy or assumptions — not directly comparable without scenario caveats.

[F] IEA, Net Zero by 2050 (2021); Net Zero Roadmap 2023 Update

The Shift Has Reached Shipping

In March 2026 the IEA began counting methanol- and ammonia-fuelled ships inside the clean energy technology market.

IEA ETP-2026 · CLEAN ENERGY TECHNOLOGY MARKET COMPONENTS

■ Electric cars

■ Batteries (EV & stationary)

■ Solar PV modules

■ Wind turbines

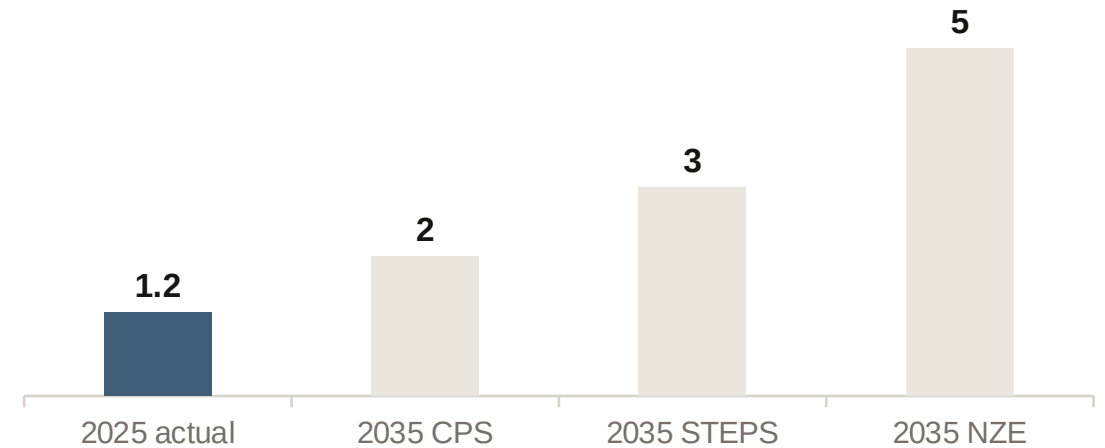
■ Heat pumps

■ Electrolysers

■ **Alternative propulsion ships**

IEA definition: “methanol- and ammonia-fuelled ships”

CLEAN ENERGY TECHNOLOGY MARKET VALUE · USD TRILLION



[F]

Three years ago these vessels sat in the “under development” column. The reclassification we have been debating has now happened inside our own sector.

CONTEXT · A market that grew ~20% a year to USD 1.2 trillion now counts our ships inside it — but being counted is not being declared mature.

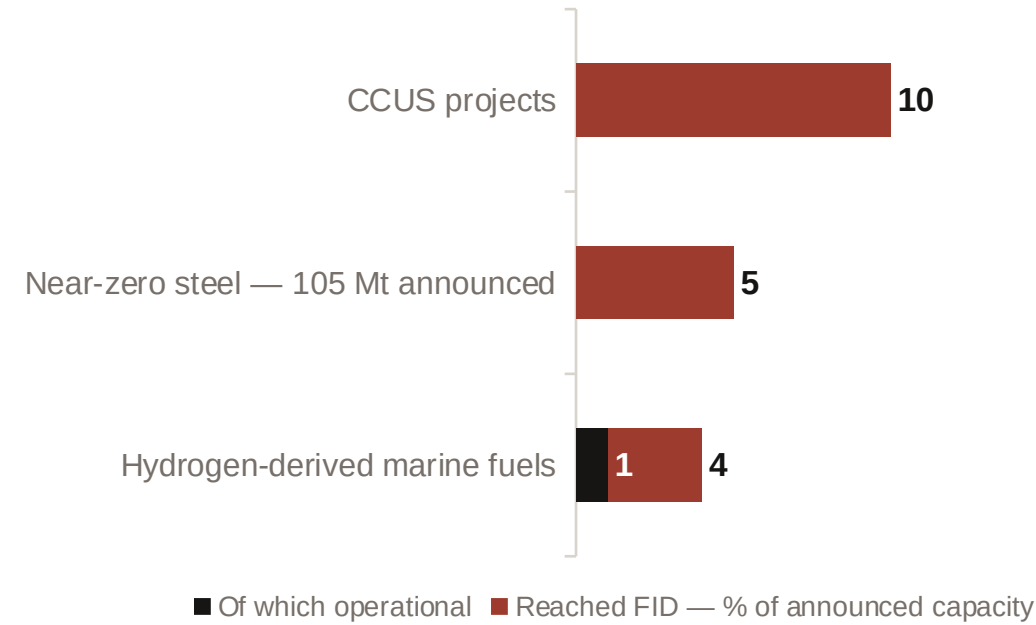
CPS / STEPS / NZE are IEA scenarios, not a single forecast. 2035 values rounded; the NZE figure exceeds USD 5 trillion.

[F] IEA, Energy Technology Perspectives 2026 (26 Mar 2026)

The Residual Burden

The market is growing — but most of the growth comes from routes that are already mature.

ANNOUNCED CAPACITY REACHING FINAL INVESTMENT DECISION · %



WHERE THE GROWTH ACTUALLY COMES FROM

60%

of low-emissions fuel market growth to 2035 comes from mature biofuels — biomethane, bioethanol, biodiesel.

Low-emissions hydrogen investment	~USD 8 bn in 2025 · +80% y/y
CCUS annual investment	15× higher than 2020 · >USD 5 bn
Early-stage technologies	“unlikely to reach significant market share within a decade”

[F] Announcement is intent. FID is readiness. The Supply Readiness Deficit is precisely the distance between the two.

CONTEXT · USD 215 bn → ~390 bn by 2035, yet hydrogen-based fuels drive little of that growth — announcement is intent, a final investment decision is readiness.

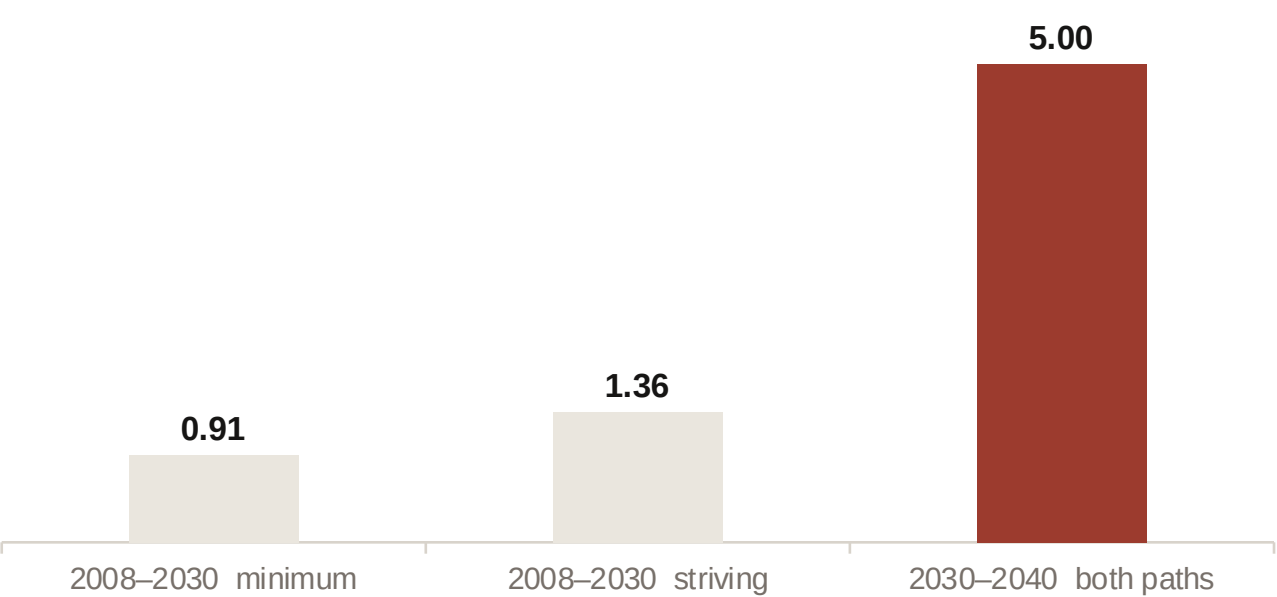
FID shares reflect the IEA project database as of the ETP-2026 cut-off (November 2025).

[F] IEA, Energy Technology Perspectives 2026

IMO Targets Imply a 3.7–5.5× Acceleration

Not a forecast. A transparent reading of the time structure embedded in the official targets.

REQUIRED PACE OF REDUCTION · PERCENTAGE POINTS PER YEAR



[A] AUTHOR CALCULATION

Minimum path

20 %-pt / 22 yr = 0.91

50 %-pt / 10 yr = 5.00

→ 5.5×

Striving path

30 %-pt / 22 yr = 1.36

50 %-pt / 10 yr = 5.00

→ 3.7×

[H] You can check this arithmetic from your seat. That is the point: if complementary assets are not in place before 2030, the physical catch-up afterwards is not available.

CONTEXT · Both paths share the same second leg — 50 points in ten years; whatever the 2020s achieve, the 2030s must run 3.7–5.5× faster.

Simple linear slope comparison between IMO checkpoints on total GHG. Not a forecast; annual emissions are not assumed linear.

[A] Author calculation on IMO Res. MEPC.377(80)

Readiness ≠ Availability

Fleet capability, regulatory requirement and actual consumption are three different numbers.

TAG	VALUE		DEFINITION	REFERENCE
[F]	50	Mtoe	Capacity of the alternative-fuelled fleet to burn low-GHG fuels ≈ 18% of fleet energy · LNG largest, then methanol; NH ₃ and H ₂ negligible to 2030	2030
[F]	25	Mtoe	Volume needed to meet the IMO 2030 base target All-sector supply 2030: 70–100 Mtoe — shipping needs up to a third of it	2030
[F]	1	Mtoe	Low-GHG fuel actually consumed ≈ 1/25 of the 2030 required volume	TODAY
BOTTLENECK	AVAILABILITY		PRICE	PORT
				CERTIFICATION

50 and 25 are 2030 values; 1 is a current value. Do not read as a same-year supply ratio.

[F] DNV, Maritime Forecast to 2050 (11 Sep 2025)

The Regulatory Stack

If we do not separate what is law from what is still under negotiation, the whole strategy collapses.

INSTRUMENT	LEGAL STATUS	SCOPE	APPLIES FROM
2023 IMO GHG Strategy	ADOPTED	Goal levels only	2023
2024 LCA Guidelines MEPC.391(81)	ADOPTED	WtW accounting boundary	2024
FuelEU Maritime	IN FORCE	WtW intensity: -2% (2025) → -80% (2050) vs 91.16 g/MJ	2025
EU ETS — maritime	IN FORCE	Surrender 40→70→100% (2024–26); CH ₄ and N ₂ O included	2026 emissions
North-East Atlantic ECA	ADOPTED	SO _x , PM, NO _x	1 Sep 2027
IMO Net-Zero Framework	PENDING ADOPTION	GHG fuel intensity + pricing	ES.2 resumes 4 Dec 2026
Direction was fixed in 2024. Open items — timing and stringency — defer supply-side investment, widening the gap between required and delivered low-GHG energy.			

The 4 December 2026 resumption is subject to confirmation by MEPC 85 (30 Nov – 3 Dec 2026). The October 2025 vote fell 49 for, 57 against, 21 abstained.

[F] IMO MEPC 84 summary; EC; Reg. (EU) 2023/1805

PART I · 08 / 32

II

The Supply Readiness Deficit

Readiness is not one number. It is physical supply, timing and verifiability — observed separately, never inferred from one another.

- MODEL VSR = $q \times \tau \times c$, and SRD is what remains
- PORTS Busan and Ulsan as coordinates, not grades
- MARKET Orders and deliveries record different things

PART I

PART II

PART III

PART IV

COMPRESSION

DEFICIT

ARCHITECTURE

SOVEREIGNTY

What changed?

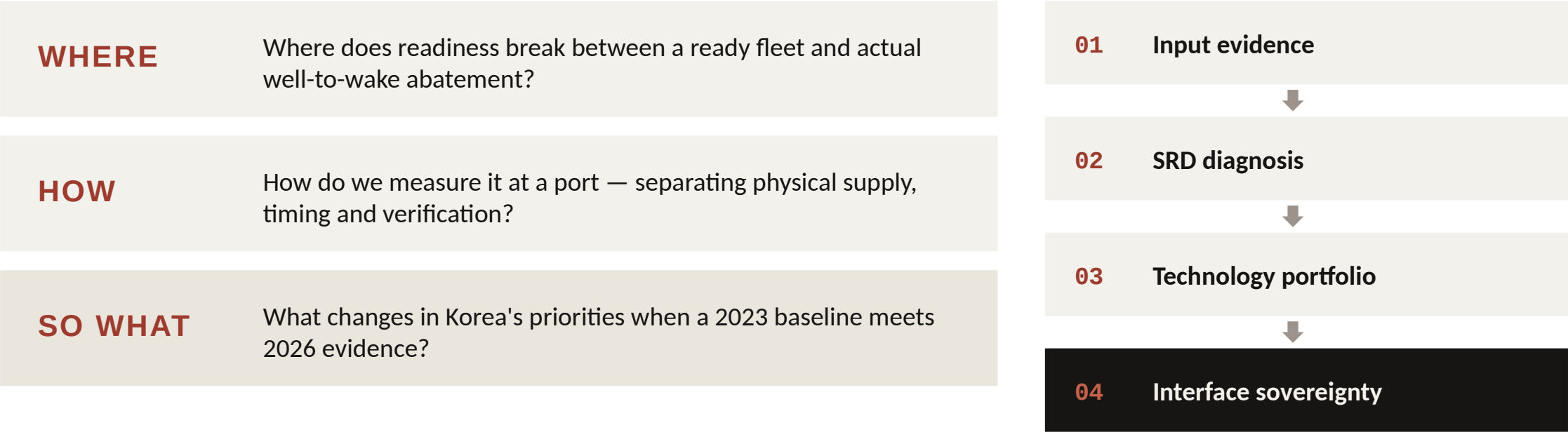
Where does it break?

What do we build?

What do we own?

Three Questions, One Measurement Language

Not which fuel wins — but where readiness breaks, how to measure it, and what that changes.



[H] Contribution
The policy question shifts from “which fuel will win” to “who will own the interfaces every fuel pathway must pass through.”

CONTEXT • Before any score is quoted the unit of analysis is fixed — vessel type, fuel, period, berth — and every number carries one of the five evidence tags.

Basis: regulatory instruments, market statistics, port operating records, and a 2023 expert panel of 45 specialists.

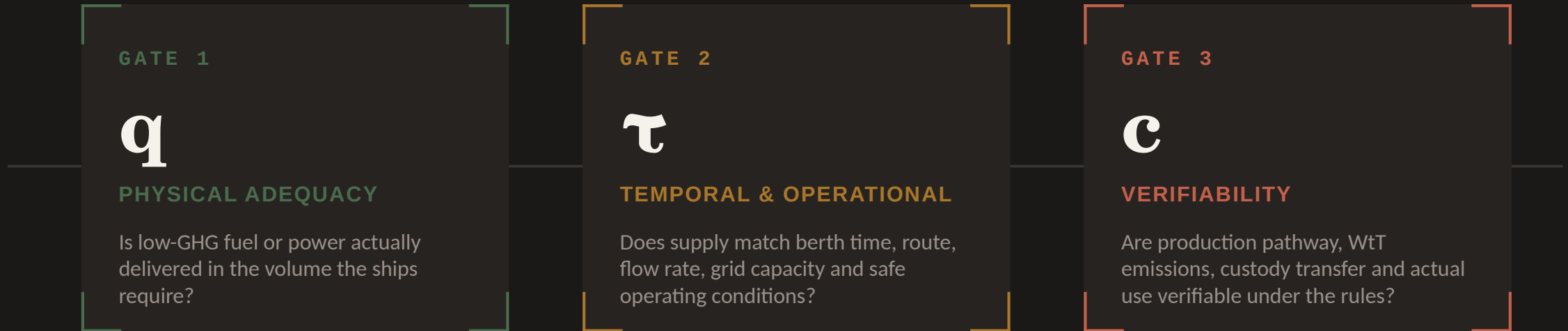
Ahn (2023); Ahn (2026), Korea Maritime Institute

SRD: The Missing Interface

Low-GHG energy that exists but arrives at the wrong time, or cannot be verified, does not count as abatement.

PRODUCTION

REGULATORY ACCOUNT



- // Contract readiness (κ) and financeability (f) are leading indicators of future supply — they are not physical supply and are kept separate from q .
- // System boundary runs from production through transport, port storage, bunkering, shipboard use and emissions measurement to regulatory reporting.

CONTEXT • The cumulative logic is borrowed from law: EU Regulation 2023/1184 already demands additionality plus temporal and geographic correlation for renewable hydrogen.

Do not infer that missing certification implies zero physical supply. The three variables are observed independently.

[M] Ahn (2026), Korea Maritime Institute

A Reproducible VSR–SRD Formulation

The purpose is not a precise-looking number. It is to isolate which interface is actually binding.

$$q(t) = \min [1 , S_v(t) / D_r(t)]$$

verified low-GHG energy delivered ÷ energy required

$$VSR(t) = q(t) \times \tau(t) \times c(t)$$

$0 \leq VSR \leq 1$

$$SRD(t) = 1 - VSR(t)$$

higher value = larger deficit

$$\dot{q}/q = \dot{S}_v/S_v - \dot{D}_r/D_r$$

when demand outruns supply, the gap widens

EQUATION → EVIDENCE

q	Bunker delivery notes, mass flow meters, metered MWh
τ	Berth records, hourly grid emission factors, flow-rate logs
c	Certificates, BDN metadata, chain-of-custody records
κ · f	PPAs and offtake contracts — leading indicators only

[M] Never insert an arbitrary 0 or 1 for an unmeasured q, τ or c. Fix the unit of analysis — vessel type, fuel, period, berth — before any port-level score is quoted. Original paper: CSR = α·β·γ, srd = 1 – CSR.

[M] Ahn (2026), eqs. (1)–(6)

Evidence Protocol: Make Every Number Auditable

In a contested field, trust comes from showing where a number came from — instantly.

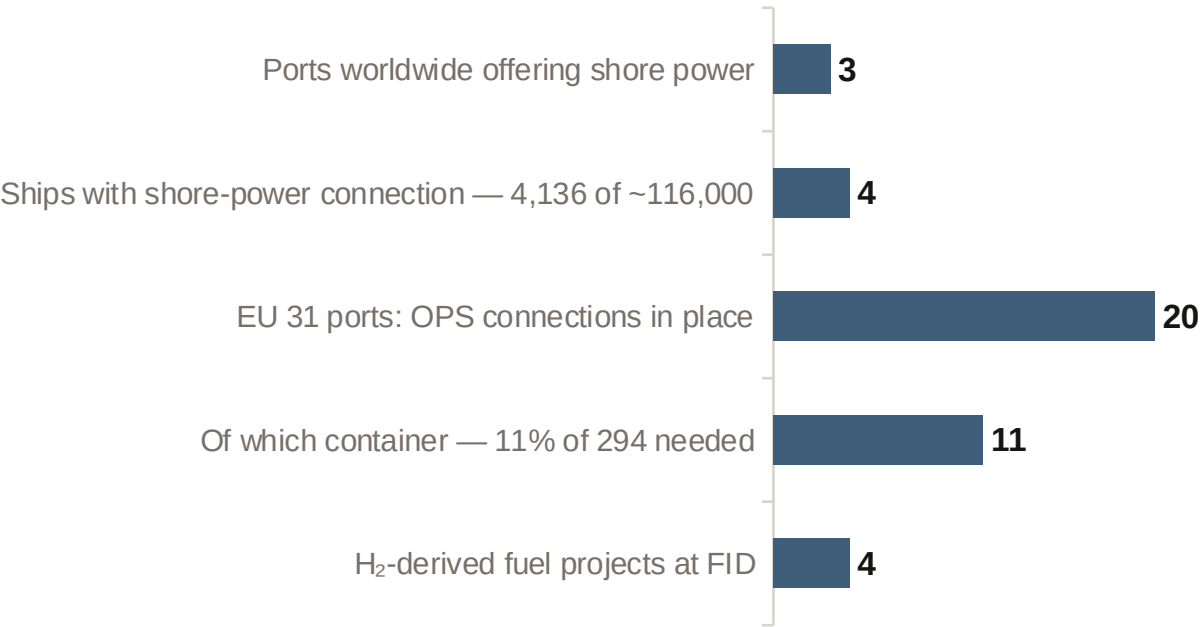
[F]	VERIFIED FACT	Official source, legislation, or direct institutional publication	MANDATORY METADATA - Cut-off date - Geographic / vessel / route scope - Unit and denominator - Source URL and conversion formula - Uncertainty range - Scheduled update date
[A]	AUTHOR CALCULATION	Reproducible from published inputs and a stated formula	
[M]	MODEL OUTPUT	Depends on assumptions, boundaries and sensitivity	
[E]	EXPERT ELICITATION	Delphi result, with survey date, panel and dispersion stated	
[H]	STRATEGIC HYPOTHESIS	Interpretation or policy proposition still to be tested	

Missing data is recorded as [NA] with a range where possible. “No data” is never treated as “readiness = 0”.

Global Supply Readiness: A Structural Deficit

Before Korean ports: this is not a national failing. It is the shape of an early transition.

READINESS AGAINST REQUIREMENT · %



BUNKERING FACILITIES WORLDWIDE

106	bio-LNG / e-LNG
1	ammonia
1	hydrogen

Container newbuild shore-power fitting rose from 30% (2022) to 63% (2025) — the bottleneck is moving shoreside.

[F] There is exactly one ammonia bunkering facility in the world. The figures you are about to see for Busan and Ulsan are coordinates, not failing grades.

CONTEXT · AFIR demands shore power at 90% of relevant calls by 2030; cruise stands at 38% — the gap is global: read Busan and Ulsan as coordinates.

The EU figure is a 31-port sample selected by activity and data availability, not the full EU network.

[F] DNV (2026); DNV for T&E (2025); IEA (2025)

Busan: Progress Confirmed, Attributes Unmeasured

The problem is not the absence of a plan. It is that delivered MWh, timing and carbon attributes are not traceable to vessel use.

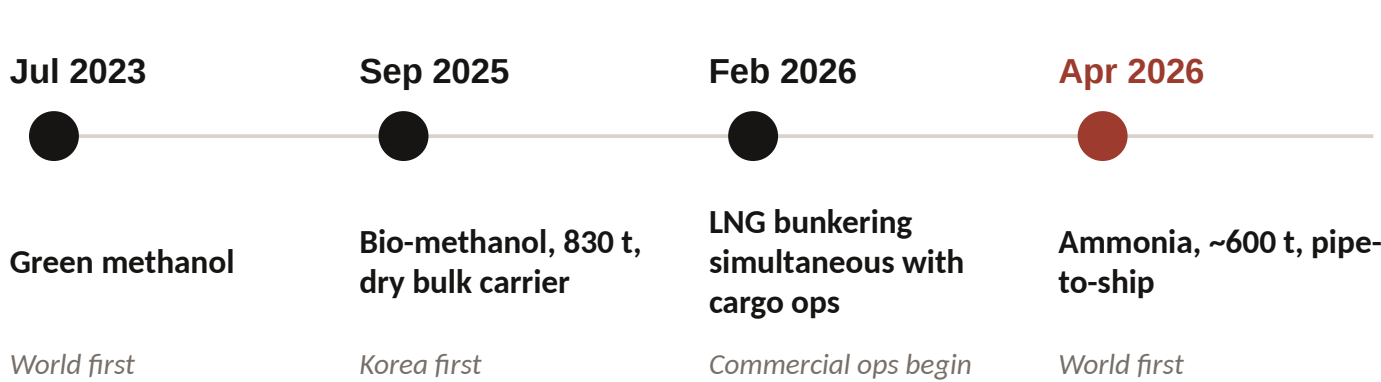
CONFIRMED — CONTRACT READINESS κ	TO BE EVIDENCED
Sep 2025 BPA – SK Innovation RE100 agreement: rooftop and idle-land solar, supply via PPA	NA Annual MWh actually delivered
Nov 2025 Busan (Gangseo) designated a first-round Distributed Energy Special District	NA Allocation to OPS berths and terminals
2027 → 2030 District LFP storage scaling 250 → 500 MWh, with “port” named as an eligible offtaker	NA Metering boundary
	NA Hourly attribute of the power supplied
	NA WtW verification method
	q NA MWh unpublished τ NA no hourly factor c NA no WtW scheme
	NA = unobserved, not zero — each gate ships with a data request

[F] Korea does not publish hourly grid emission factors. There is no institutional basis on which τ could be measured at all — which is why recommendation one is a data mandate, not a capital programme.

CONTEXT • Direct power trading now covers ~49.9 km²; the grid factor is yearly but still annual-average — and shore-power use has run below 1% of calls.

Ulsan: Four Firsts, One Missing Metric

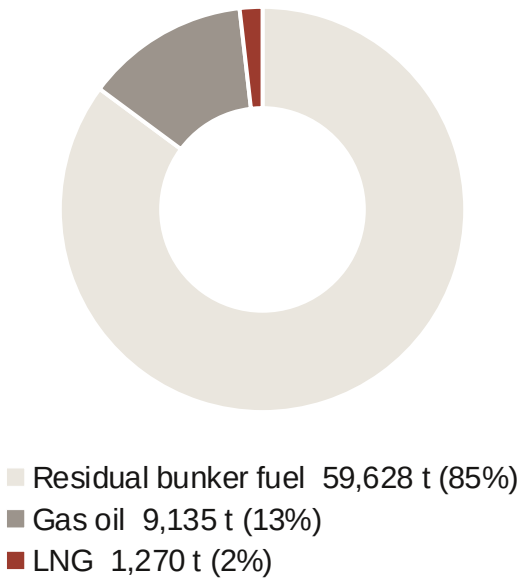
“Can it be done?” has been answered. “How often, how much, with what WtW attribute?” has not.



The only port in the world to have supplied LNG, methanol and ammonia.

And in the same month as the ammonia first, 85% of the fuel Ulsan actually delivered was residual bunker fuel. Both statements are true; they measure different things.

ULSAN BUNKERING MIX · APRIL 2026



[A]

Alternative fuels $\approx 0.29\%$ of supplied energy $\rightarrow q \approx 0.003$; $c \approx 0$ (no WtW certification scheme in national law)

Basis: 664 international port calls in April 2026 (–16.6% y/y) — 70,033 t of fuel supplied in total

$q \approx 0.003$ measured — near-empty	τ not yet measured	$c \approx 0$ no WtW certification
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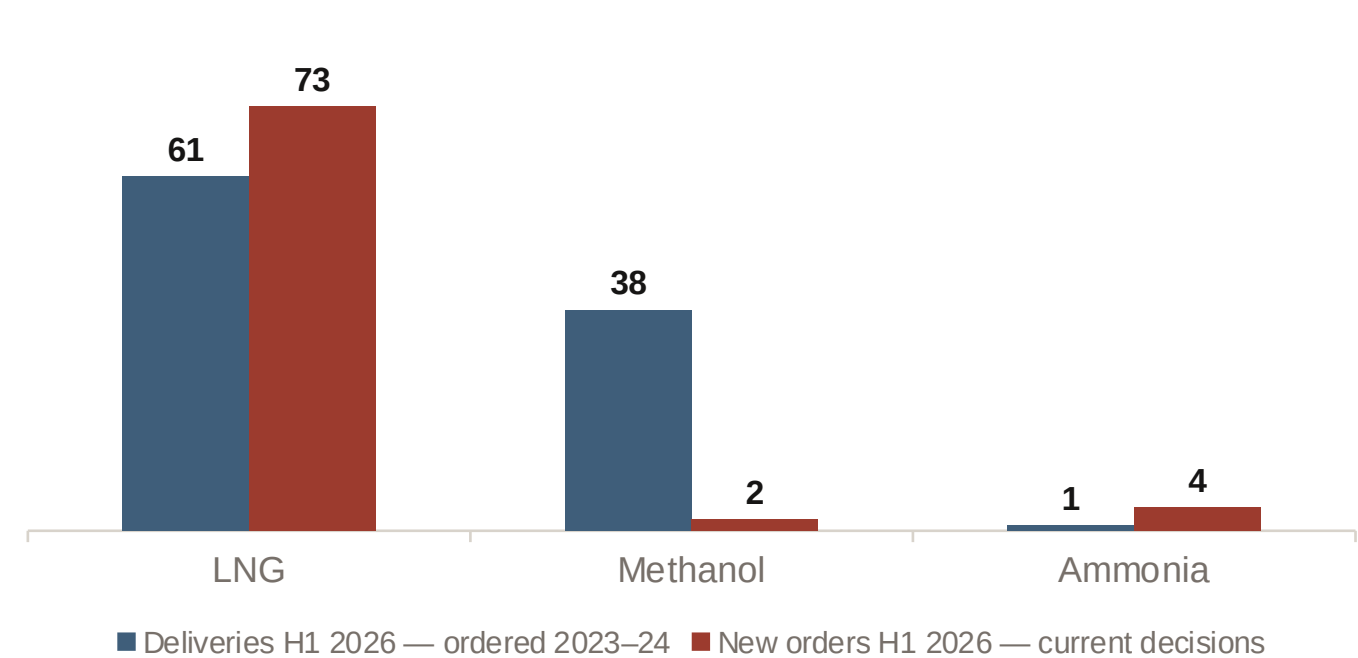
Converted on an energy basis, not by mass. Quarterly annualisation ignores seasonal variation. April 2026 volumes were 59,628 t residual, 9,135 t gas oil and 1,270 t LNG across 664 international calls (–16.6% y/y); $\pm 30\%$ on the annualised volume keeps q within 0.002–0.004.

[F] MOF / Ulsan Customs / UPA (2026)

Orderbook Signals: Two Numbers, One Market

Methanol orders collapsed to 2. Methanol deliveries in the same half-year were 38.

ONE HALF-YEAR, TWO QUANTITIES · NUMBER OF VESSELS



WHERE THE 137 NEW ORDERS WENT · H1 2026



Say “consistent with”, never “proved by”. Orders, operating fleet and fuel consumption are distinct quantities. The orderbook still holds 313 methanol ships against 663 LNG; the fleet is 95% conventional by tonnage while 35% of orderbook GT is alternative-ready.

137

ordered in H1 2026 — down from 155 in H1 2025 (–11.6% y/y)

149 → 2

methanol orders: 149 (FY 2024) → 40 (H1 2025) → 2 (H1 2026)

2,403

total newbuild orders in 2025, against 4,405 in 2024

[H]

Fuel availability is one contributing factor — not a verdict. Orders price financial decisions; deliveries record the physical fleet.



Technology Architecture

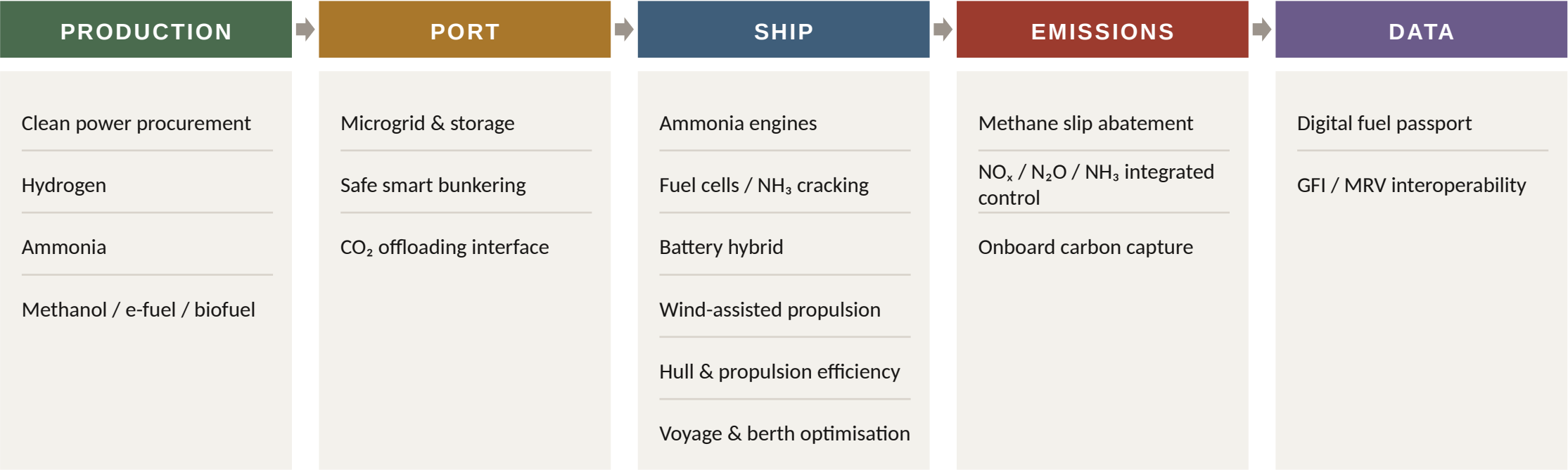
The unit of assessment is never the technology alone. It is the technology, its complementary asset, and its means of verification.

- TAXONOMY 17 technologies arranged on a well-to-wake basis
- ACCOUNTING One molecule of N₂O, three lawful answers
- EVIDENCE The 2023 expert baseline, re-read against 2026



A WtW Interface-Based Technology Architecture

The unit of assessment is not the technology — it is technology plus its complementary asset plus its means of verification.



General pre-combustion capture is removed from the 17-item core taxonomy: it does not separate carbon from ammonia. On-board reforming and methane pyrolysis are managed separately as long-horizon options.

CONTEXT • The 17 items come from Ahn (2023); what is new is promoting port, safety, data and certification from background conditions to technology groups.

Each layer is governed by a different instrument: LCA Guidelines, AFIR/FuelEU, IGF/IGC, MEPC.414(84), SFCS and the GFI Registry.

Ahn (2023), extended on an interface basis

Ammonia Engines: Deployment ≠ Ecosystem Readiness

A specific two-stroke high-pressure architecture entered commercial service. The value chain did not.

READINESS BY LAYER · INDICATIVE		
Engine type approval	COMPLETE	TAT / FAT completed Jan 2026, HHI-EMD, witnessed by LR
Vessel integration & sea trials	COMPLETE	Load response and efficiency on par with equivalent diesel X-Engines
Fuel supply chain	PARTIAL	One ammonia bunkering facility worldwide
Port safety framework	PARTIAL	Ship-side rules approved; port-side SIMOPS at national discretion
Real-world emissions verification	MISSING	No agreed international measurement standard for ammonia N ₂ O

Jan 2026	Apr 2026	Jun 2026	Jul 2026	~5%
TAT / FAT	Ulsan bunkering	ANTWERPEN delivered	Arlon, 2nd of 4	pilot fuel at full load

ANTWERPEN and Ulsan: Who Owns What?

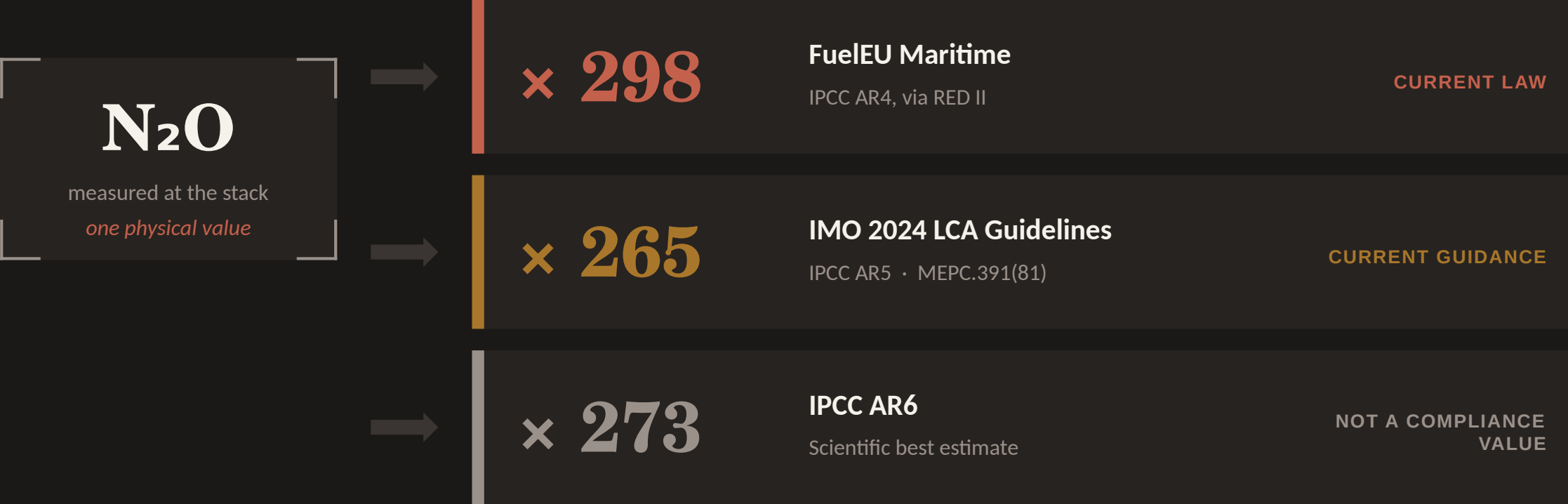
Participating in the division of labour is not the same as holding the rights it generates.

ASSET	HELD TODAY BY	TARGET STRUCTURE	INSTRUMENT
Engine design IP	WinGD (Switzerland)	Licensed — unchanged	Licence terms
Manufacturing & integration	HHI-EMD / HD HHI (Korea)	Retained	—
Sea-trial and test-bed data	Shared, undefined	Joint use right, named parties	Consortium agreement
Bunkering SOP	Ulsan / Lotte Fine Chemical	Codified and exportable	Port standard
Real-world emissions data	Undefined	Co-owned, third-party verifiable	Grant agreement
Right to co-propose standards	Not secured	Named co-proposer at IMO / ISO	Project contract

Technology sovereignty is not a flag on a hull. It is who owns the data, who defines the test protocol, and who is named when the standard is proposed.

One Molecule, Three Accounts

The same engine, the same measurement — and three different answers in CO₂-equivalent.



[A] An identical measured N₂O mass is booked ~12.5% higher under FuelEU than under the IMO guidelines — and the EU intends to migrate to AR5. The contest is not chemistry. It is accounting.

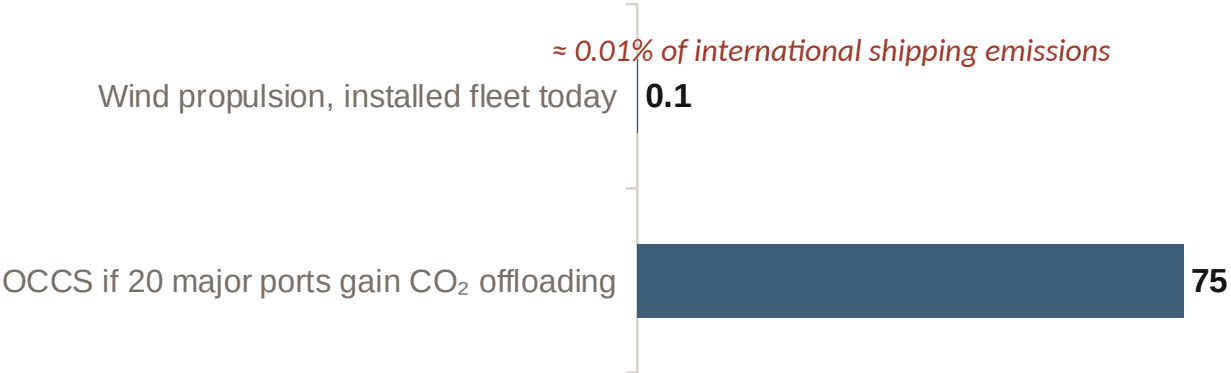
Every figure in this deck states which coefficient set was applied. AR6 273 appears only as a scientific footnote. CH₄ moves too: 25 (AR4) vs 28 (AR5); the EC intends to migrate FuelEU to AR5 in line with Reg. (EU) 2023/957. [F] Reg. (EU) 2023/1805; IMO Res. MEPC.391(81); IPCC AR6

PART III · 20 / 32

Portfolio Complements: WAPS and OCCS

The window of opportunity is not open or shut. It migrates — from hardware to integration, data and regulatory recognition.

ANNUAL ABATEMENT · MILLION TONNES CO₂ PER YEAR



Wind's 100-vessel milestone is real. So is the fact that it abates one hundredth of one percent. The frontier is not the sail — it is the measurement.

[F]

OCCS at 20 ports ≈ 9% of shipping emissions — the upside is large and entirely infrastructure-gated.

WIND-ASSISTED PROPULSION

Commercial hardware · open MRV window
Frontier: separated in-service savings measurement, large-ship structural integration, MRV recognised under CII / FuelEU / GFI.
100+ ships · 230+ systems · 5M+ DWT · 5–20% savings per retrofit · ~200 ships by mid-2027

ONBOARD CARBON CAPTURE

Mature technology · conditional infrastructure
Accounting, verification and certification will not conclude before MEPC 86 (autumn 2027).
TRL 7–8 · 15+ pilots · fuel penalty 9–30% (DNV model: 75% capture, 20% penalty)

CONTEXT · Wind hardware is commercial — its frontier is measurement; capture is mature — its frontier is ports and accounting. The device is not the bottleneck.

The 20-port figure is a DNV model scenario, not an observation. Do not double-count wedges.

[F] IWSA (Jun 2026); ICS OCCS Review (Jun 2026); DNV (2025)

A Revised Delphi-to-Decision Framework

A low domestic capability score is not, by itself, a high priority.

STEP 1

2023 BASELINE

[E]

L, M, D, E from a two-round Delphi of 45 experts (21 domestic, 24 European), Jul–Aug 2023. Medians, IQR and consensus published.

STEP 2

2026 OVERLAY

[F]

Type approvals, commercial deliveries, regulatory application, demonstration volumes, standards development, patent and supply-chain shifts.

STEP 3

DECISION RULE

[M]

$$P = V \times (1 - L/100), \quad V = (M+E)/2$$
$$P^* = P \times A \times I \times R \times C$$

A absorptive capacity · I interface centrality · R regulatory urgency · C evidence confidence

THE MISREADING THIS CORRECTS

P is a gap index, not a ranking.

P rises as domestic capability L falls. Pre-combustion capture scored P = 1.902 and sixth place in 2023 because L was 58.2 — not because it was close to breakthrough.

D is not dropped from the rule. It operates as a stage gate: $D \geq 4.2$ with no dominant design locked in → exploratory joint research.

CONTEXT • 45 experts — 21 Korean, 24 European — in two rounds, Jul–Aug 2023; L doubles as exchange value and absorptive capacity, D decides if the window is open.

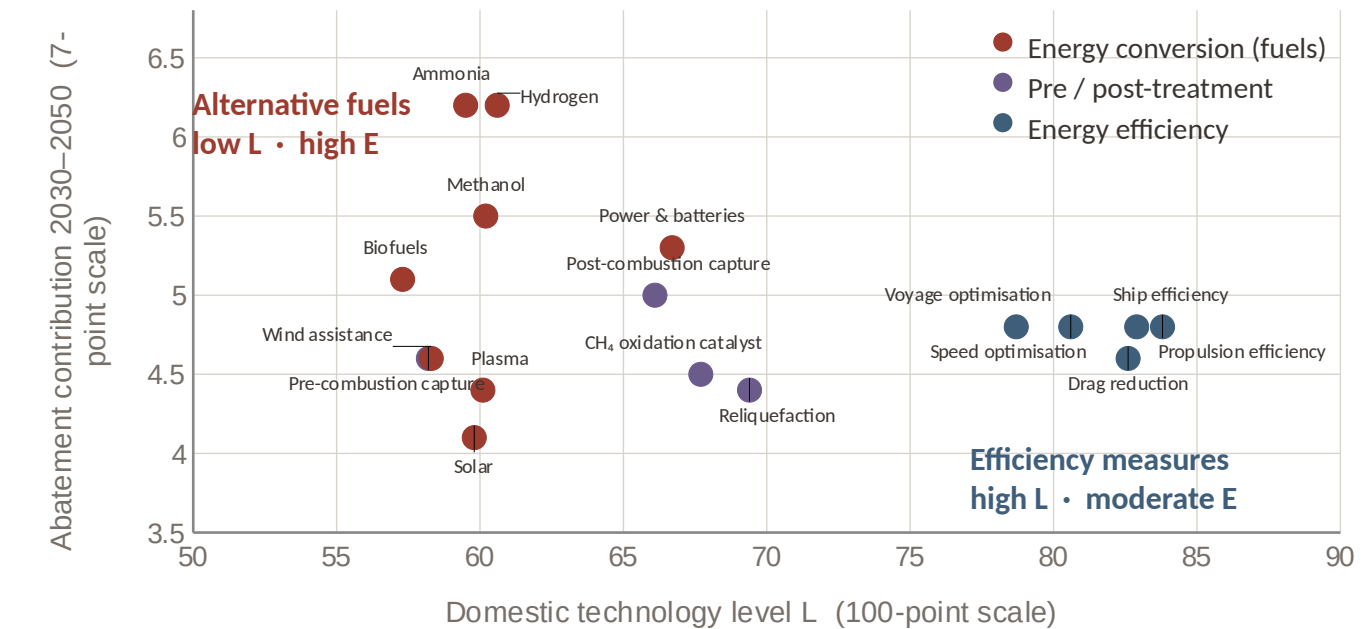
The correction factors are proposed as a frame. P* is not recomputed here — that requires an updated panel and dataset.

[E] Ahn (2023), eq. (1); [M] correction factors proposed here

The 2023 Capability Mismatch, Re-read in 2026

Where capability is highest, future abatement value is lowest — and commercialisation has exposed the next interface.

2023 DELPHI · CAPABILITY VERSUS FUTURE VALUE · 17 TECHNOLOGY ITEMS



Energy efficiency	L = 81.7 E = 4.8
Pre/post-treatment	L = 64.3 E = 4.6
Energy conversion	L = 60.3 E = 5.3
— hydrogen & ammonia	E = 6.2

2026 EVIDENCE OVERLAY

- ➔ Jan 2026 type approval and factory acceptance
- ➔ Apr 2026 world-first ammonia bunkering
- ➔ May 2026 MSC 111 interim safety guidelines
- ➔ Jun 2026 first ocean-going delivery

COMMERCIALISATION EXPOSED THE NEXT GAPS

Real N₂O and NH₃ emissions · toxicity and SIMOPS safety · fuel chain-of-custody · port energy and data interoperability

[H] The gap did not close with engine capability. Convert strong ship and efficiency capability into exchange assets through interface demonstrations — do not “drop efficiency, buy fuels”.

CONTEXT · Korea's strength is real where value is lowest — 73 of 216 efficiency mentions (33.8%), ahead of Europe on drag reduction: assets to trade, not abandon.

Points are the 2023 expert baseline and are not moved. 2026 events are shown as an overlay, not as a rescoring.

[E] Ahn (2023), Tables 2 and 6; [F] 2026 milestones

IV

Interface Sovereignty

Not autarky. The capacity to negotiate with assets that others cannot do without — and with the instruments that judge them.

- SELECTION Four gates, and what fails them is named
- FRONTIERS Emissions integrity · safety · MRV · nuclear interface
- SEQUENCE Measure, demonstrate, standardise, scale

PART I

PART II

PART III

PART IV

COMPRESSION

What changed?

DEFICIT

Where does it break?

ARCHITECTURE

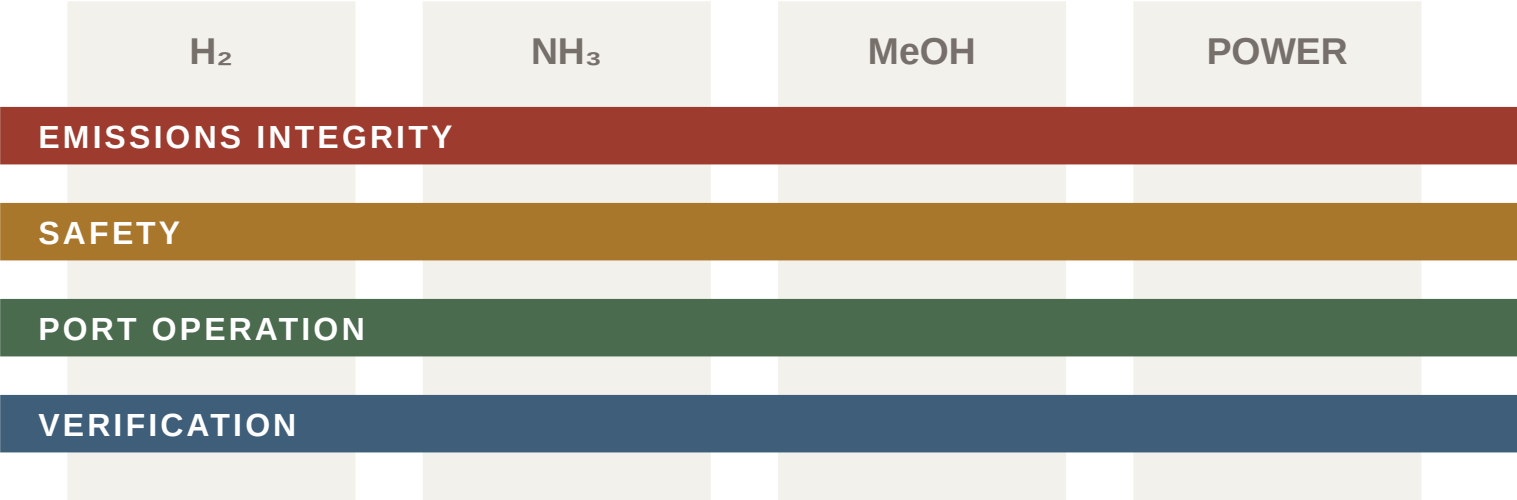
What do we build?

SOVEREIGNTY

What do we own?

Technology Sovereignty as Interface Sovereignty

Not autarky — the capacity to negotiate with assets that others cannot do without.



Vertical silos are fuel bets. Horizontal bars are what every bet must pass through.

“Own the critical interfaces, negotiate the rest.”
Horizon Europe association, signed 17 Jul 2025 — Pillar II only (ERC excluded) · runs to 2027 · FP10 separate

WHAT KOREA MUST HOLD — FOUR ASSETS

01 • REAL-SHIP DATA

Build, integration and sea-trial data from large vessels in service

02 • PORT PROTOCOLS

Bunkering, shore power, CO₂ offloading and SIMOPS records and test protocols

03 • VERIFICATION

Real emissions, WtW data models, third-party verification — interoperable standards

04 • RIGHTS ON PAPER

Foreground IP, joint patents, data-use rights, standard co-proposal rights, testbed continuity

CONTEXT • Not autarky (Edler et al. 2023): never depending on another's unilateral decision — cooperation becomes exchange, and you prove what you bring first.

Association confers eligibility, not intellectual property. Rights are secured per project, in the consortium and grant agreements.

[F] Agreement EU–Republic of Korea, OJ L 2025/1532

Selecting Frontier Technologies: Four Gates

The gates decide where Korea commits demonstration, standards and IP effort first — and what fails is named.



BONUS CRITERIA cross-fuel applicability · exportability of the resulting standard or service

PASS ALL FOUR ► THE FRONTIER PORTFOLIO — SLIDES 26-29
01 Ammonia emissions integrity · 02 Safety and smart bunkering · 03 Digital fuel passport · 04 Nuclear-ready interface

EXCLUDED **General pre-combustion carbon capture**
Passes G2, fails G1 and G4 — separated CO₂ still needs offloading and storage; maritime TRL 4-5. Detail on slide 30.

CONTEXT • The gates descend from windows of opportunity, absorptive capacity and sovereignty — and a selection rule only proves itself when something fails it.

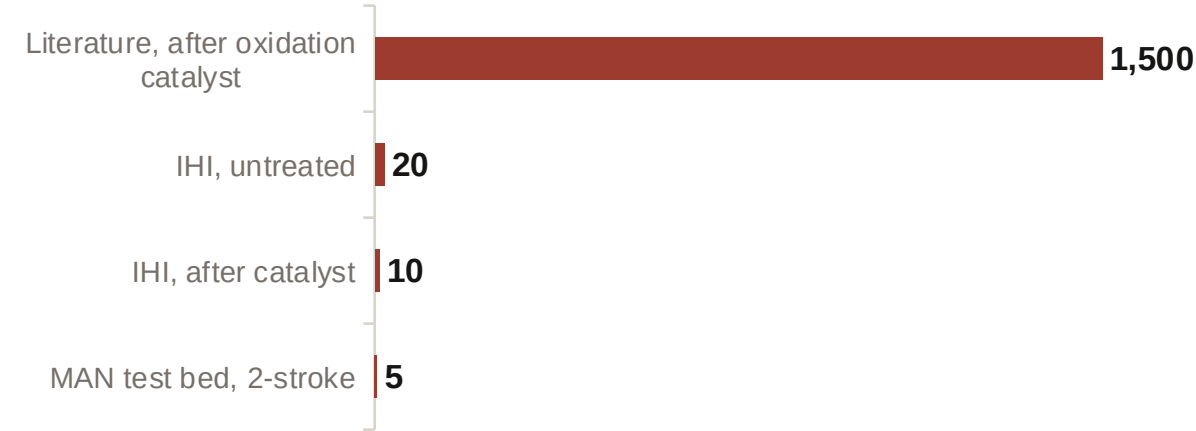
Technologies whose safety or environmental performance cannot be verified are withheld at source, whatever their promise.

[M] Selection framework proposed here

Ammonia Emissions Integrity Stack

The engine is commercial. The instrument that judges its climate performance is not yet built.

REPORTED N₂O CONCENTRATION · PPM · LINEAR SCALE



Three orders of magnitude between the manufacturer test bed and the reported literature maximum.



Physical constraint: peak N₂O formation occurs at 200–400°C — exactly the exhaust temperature range of low-speed marine engines. A diesel oxidation catalyst promotes its formation.

Korean asset: test-bed data at HHI-EMD, sea-trial data from ANTWERPEN and bunkering data from Ulsan — the only country able to link all three layers.

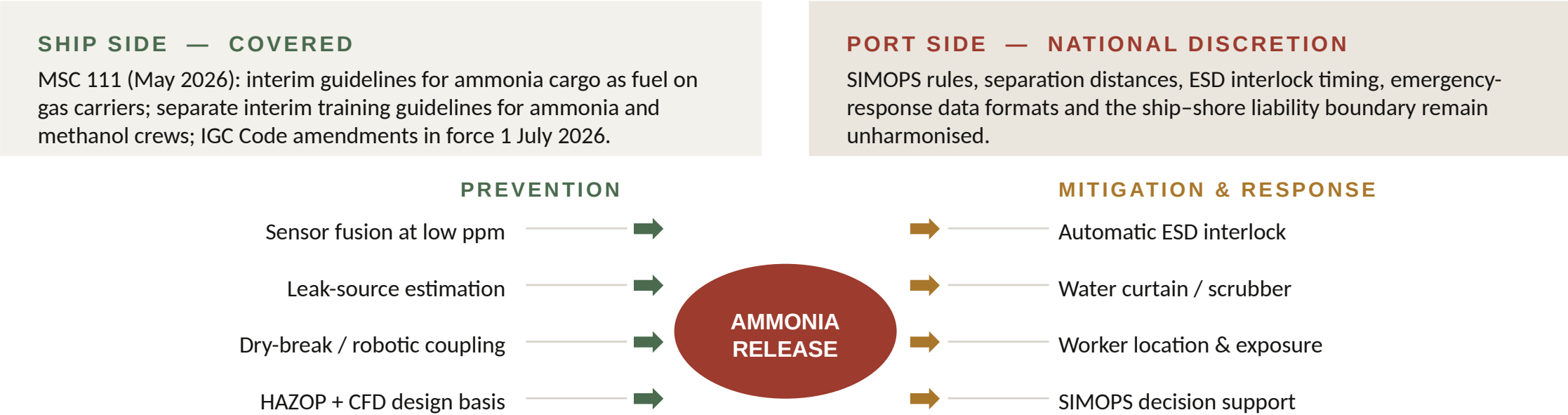
AND NO INSTRUMENT TO ADJUDICATE IT

- MEPC 84 adopted CH₄/N₂O measurement guidelines — MEPC.414(84), ELM MEPC.415(84), CEMS — and stated the framework still needs development to cover ammonia.
- The EC's FuelEU methane-slip guidance explicitly excludes N₂O measurement procedures for want of data.

Do not generalise: N₂O varies by engine architecture, load and aftertreatment. 1,500 ppm is a specific reported condition, not a representative value. LPDF tests report N₂O 1.12→1.75 g/kWh and NH₃ 31→143 g/kWh as ammonia rises 50→90%; the MMMCZCS figures are proposals, not law.

Ammonia Safety and Smart Bunkering Intelligence

Ship-side rules filled quickly in 2026. Port-side rules did not.



DIGITAL DATA LAYER

dispersion digital twin · detection time · safe-shutdown time · exposure-minutes · false-alarm rate

Korean asset: provisional registration standard for ammonia bunkering operators (2025) and the world's first pipe-to-ship demonstration (Apr 2026).

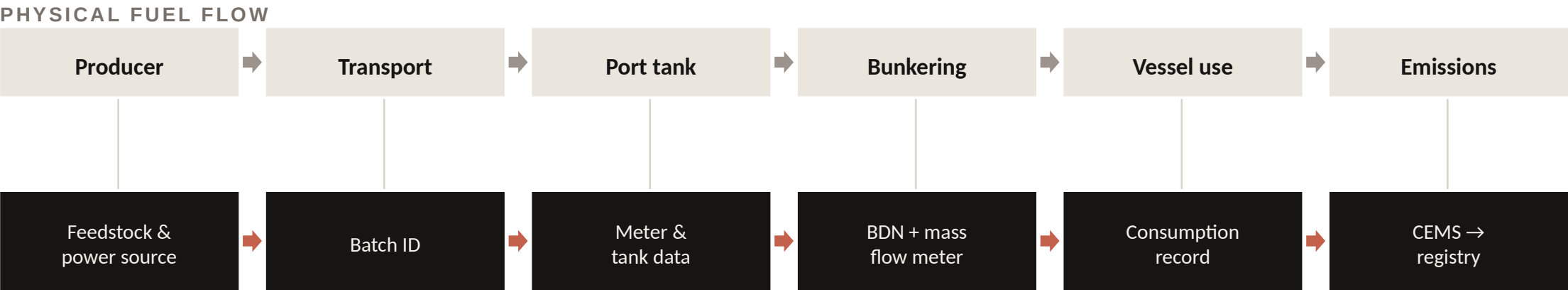
CONTEXT • The ship-side rules were drafted by Lloyd's Register, Belgian FPS Mobility and EXMAR — operating records became regulation; the port side still waits.

Occupational exposure references (≈20 ppm / 8 h, 50 ppm / 15 min, 300 ppm IDLH context) are study reference values, not a single international legal standard.

[F] IMO MSC 111 (May 2026); EMSA; MOF (2026)

Digital Fuel Passport and GFI-MRV Interoperability

The rules are being written right now — in a specific room, with a specific agenda.



DATA LINEAGE

THE STANDARD IS BEING SET NOW

15 Jul 2026 IMO expert workshop GHG-EW 7 on chain-of-custody models: physical segregation, mass balance, book-and-claim

In development SFCS Guidelines, Fuel Lifecycle Label, and the GFI Registry — traceability with double-counting prevention as a core principle

KOREA'S ENTRY TICKET

4 → 24

bunker supply vessels fitted with mass flow meters, 2025 pilot to 2026 — the only physical infrastructure improving c(t) today.

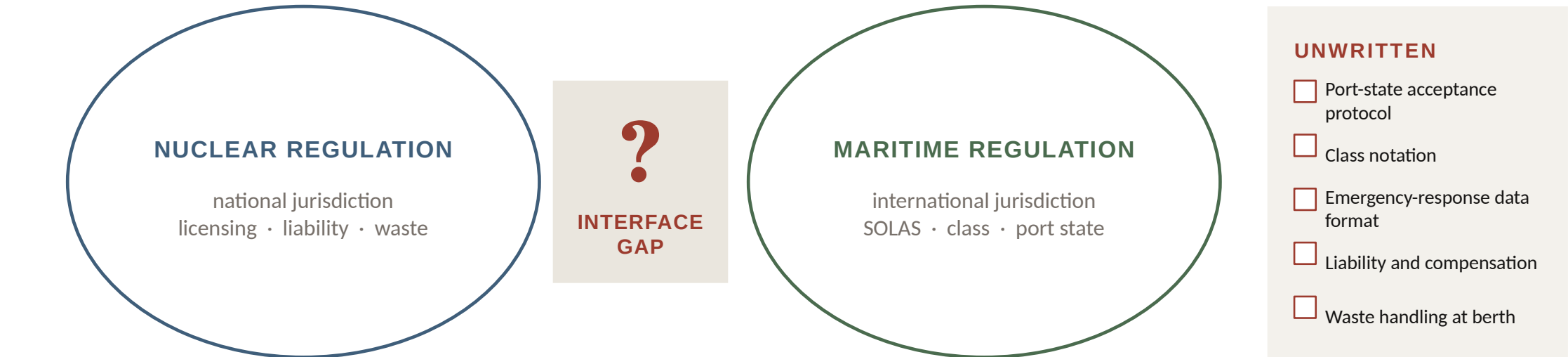
CONTEXT • If IMO is slow the de-facto standard is set elsewhere: ISO 22095-3 in drafting, Katalist live, ZEMBA owners already claiming fuel tokens.

Blockchain is an optional implementation, not a requirement. What matters is the data model, permissions, auditability and a standard API.

[F] IMO GHG-EW 7 (15 Jul 2026); ISWG-GHG 21; MOF (2025)

Nuclear-Ready Port and Flag Interface

We may never sell the reactor. The rules for letting one enter a port are unwritten — and we hold a design.



A mobile reactor crossing international waters and berthing next to a city was anticipated by neither regime.

● Jun 2025 MSC 110 opens revision of the 1981 nuclear ship regime; work assigned to the SDC sub-committee	● Jul 2025 NEMO granted IMO NGO status; IACS issues its first position paper	● Early 2026 IAEA launches the ATLAS project for maritime nuclear licensing	● Jul 2026 ABS grants AiP to KRISO, KAERI and Samsung Heavy Industries for a 15,000 TEU containership with two small modular molten salt reactors
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An AiP is neither a construction permit nor an operating licence — and the codes it was assessed against date from 1981. Commercial deployment is projected from the 2030s. Two redundant molten-salt reactors share the load, with storage absorbing surplus; Bureau Veritas expects pilots in the 2030s, commercial service around 2045.

[F] IMO MSC 110; IACS (2025); ABS / World Nuclear News (Jul 2026)

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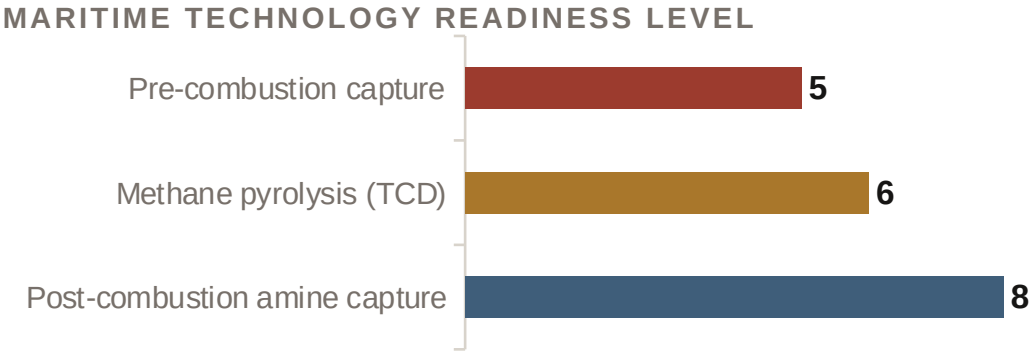
Conditional and Long-Horizon Options

Technical imagination is necessary. It cannot outrun chemistry and a mass balance.

EXCLUDED FROM THE CORE PORTFOLIO

General pre-combustion capture

It separates CO₂ from syngas — the offloading and storage burden remains. Solid carbon comes from methane pyrolysis, a different pathway entirely. Ammonia contains no carbon.



TIER 2 · CONDITIONAL

- OCCS – port CO₂ interface**
Accounting, verification and certification will not conclude before MEPC 86 (autumn 2027); MARPOL versus London Protocol jurisdiction unresolved.
- Methane slip abatement**
Reduces transition-fleet climate risk across a 663-vessel LNG orderbook. Not a long-term zero-carbon answer.
- WAPS performance assurance**
Separated in-service measurement and regulatory recognition, not the sail itself.

LONG HORIZON

- NH₃ cracking – SOFC**
Commercial systems below 1 MW against a 10–20 MW propulsion requirement. No commercially financed order confirmed as of early 2026.
- Onboard reforming + CO₂ separation**
Maritime TRL 4–5. The offloading and storage burden persists.
- Methane pyrolysis**
CH₄ → 2H₂ + C(s). Viability turns on solid-carbon quality, market and disposal, process heat, and upstream methane leakage.

Pre-combustion capture does not produce solid carbon. OCCS is not paralysed — the technology is at TRL 7–8; only the accounting rules are unfinished. Viking Energy's 2 MW SOFC retrofit has slipped; no commercially financed order confirmed (early 2026).

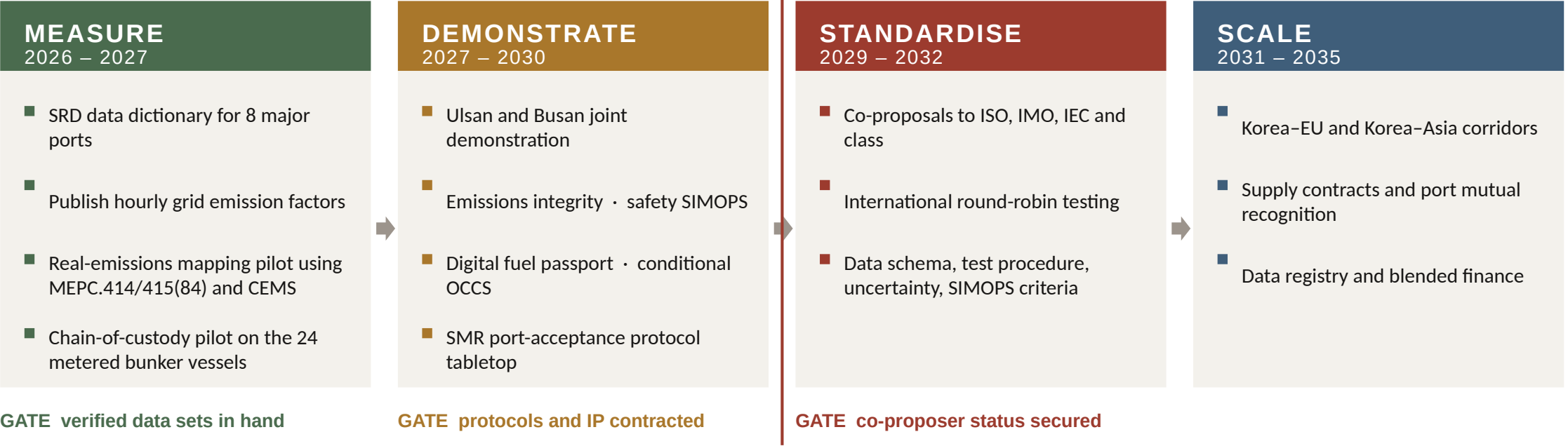
[F] ICS OCCS Review (Jun 2026); EMSA (2026); IBIA (May 2026)

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Roadmap 2026–2035

Measure, prove, standardise, scale — the purpose of public money is the verified data that makes the next standard possible.

HORIZON EUROPE ASSOCIATION EXPIRES 2027



The standardisation phase begins in 2029. The association agreement runs only to 2027. Unless FP10 participation terms are negotiated during 2026–27, that phase does not exist.

POLICY KPIS

verified data sets · test protocols · patents · standards submissions · contracted low-GHG energy

CONTEXT • The ₩320bn MOTIE programme and ₩7.36tn MOF budget already exist — redefine what they buy: verified data and standards, not berth counts.

Domestic pillars: an SRD observatory publishing q , τ and c annually; a regulatory sandbox and shared testbed; and blended finance with performance-linked support.

[F] MOTIE (2026); MOF (2025); EC (2025)

CONCLUSION

Own the Interfaces, Not a Single Fuel

EMISSIONS
INTEGRITY

SAFETY &
BUNKERING

DIGITAL
MRV

NUCLEAR-READY
INTERFACE

MEASURE transparently DEMONSTRATE jointly STANDARDISE internationally FINANCE conditionally

01

The reclassification reached shipping in 2026 — yet the targets demand a 3.7–5.5× acceleration after 2030, and 50, 25 and 1 Mtoe remain three different numbers.

02

SRD must be measured by separating q , τ and c . Missing data is never entered as zero.

03

The 2026 engine and bunkering milestones are real — and the instrument that judges them does not yet exist.

04

Sovereignty means turning ship and port data, test protocols, joint IP and co-proposal rights into negotiable assets.

Which critical interface should Korea make indispensable to the global maritime transition by 2030?

CONTEXT • One decision cannot wait: FP10 terms must be negotiated in 2026–27, or the standardisation phase beginning 2029 has no legal vehicle.